Rajdeep Gill

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Education

Bachelor of Science in Computer Engineering (Co-op) May 2026
University of Manitoba *Winnipeg, MB*

GPA: 4.45/4.5
Relevant Coursework: Data Structures and Algorithms, Parallel Computing, Computer Architecture, Operating Systems, Embedded Systems, Digital Signal Processing

Experience

Robinhood May 2025 - Sept 2025
Software Engineer Intern *Toronto, ON*

- Incoming intern for the Summer 2025 term on the Brokerage team

Biosensors Research Lab - University of Manitoba May 2024 - Dec 2024
Undergraduate Research Student *Winnipeg, MB*

- Designed and trained a convolutional neural network using Python and PyTorch to predict cell viability from 10,000+ diffraction images, achieving 85% accuracy, demonstrating potential for non-invasive cell viability assessment
- Improved particle diffraction centroid detection success rate by 17% (from 75% to 92%) using computer vision techniques with Python and OpenCV, enhancing data quality for CNN training
- Developed an interactive data visualization tool in Python with Tkinter to process filter, and display model predictions
- Engineered a SQL database architecture for experimental data management and implemented a RESTful API for easy data access and retrieval

Biosensors Research Lab - University of Manitoba May 2023 - Aug 2023
Undergraduate Research Student *Winnipeg, MB*

- Applied dimensionality reduction using Python and Scikit-learn to analyze and cluster cells based on scattering patterns
- Optimized data analysis pipeline through parallel processing and refactoring existing algorithms, achieving 2.5x speedup
- Developed an automated oscilloscope testing platform with Python and PyVISA, streamlining data collection protocols to enhance measurement speed and ensure repeatability

Projects

Collaborative Document Editor | *React, TypeScript, HTML/CSS, Next.js, Socket.io, SQL*

- Developed a collaborative document editor supporting real-time collaboration and user authentication
- Utilized Socket.io for real-time collaboration and implemented autocomplete and in-line suggestions using Google Gemini

Financial Dashboard | *React, TypeScript, HTML/CSS, Next.js, Hono.js, SQL*

- Created an expense and income tracking dashboard with user authentication and easy data visualization
- Implemented a RESTful API to handle user data and transactions, and a PostgreSQL database to store user information

Machine Learning Classification of Microplastics | *Python, PyTorch, OpenCV*

- Developed a machine learning model to classify microplastics based on their optical and dielectrophoresis features
- Created an image processing pipeline to extract features, discard low-quality images, and prepare data for training
- Paper accepted at IEEE AP-S/URSI 2025, "Machine Learning Classification of Microplastics by Integrating Optical and Dielectrophoresis Features"

Distributed Image Processing Pipeline | *C++, OpenMPI*

- Developed a distributed image processing pipeline with OpenMPI to process large images in parallel, achieving a 3.8x speedup on 4 nodes
- Implemented a custom communication protocol to ensure low latency and high throughput

Technical Skills

Programming Languages: Python, C, C++, Java, TypeScript, JavaScript
Frameworks & Libraries: PyTorch, OpenCV, OpenMPI, Tkinter,React, Node, Express, Spring Boot, Flask
Tools: Git, Jenkins, Jira, AWS, Linux, Docker, Firebase, SQL, NoSQL, Excel, REST API
Concepts: Backend Development, Frontend Development, Fullstack Development, Software Engineering, Machine Learning, Networking, CI/CD, Microservices, Infrastructure Engineering, DevOps, Automation, Agile